Posterior Sampling Time Exploration – $m = 64$ and $T = 1500$

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The overall model fitting time (with `equalTimeDist = TRUE` specified when fitting the models) for our 4 models, i.e., `fullGPfixedL`, `NNGPblockFixedL`, `NNGPsequenFixedL`, and `NNGPsequenVaryLj`, are 2.41 days, 2.41 days, 2.41 days, and 2.35 days, respectively.[1] If we do not take advantage of our tactics for evenly dispersed time points presented in Appendix B of our manuscript by specifying `equalTimeDist = FALSE` instead, we will need more than 3 months to fit the same models using the same computation resources, as $T = 1500$ is huge. This corresponds well to what we have discussed in Appendix B regarding our approaches' manifest computational acceleration in Gibbs sampler steps for temporal parameters ψ, Υ, and $\boldsymbol{\eta}_t$'s. Since $m = 64$ is quite small, there aren't any significant differences in the recorded Gibbs sampler time between our four models, as expected.

We first display the first 50 kept post-burn-in MCMC iterations' posterior sampling time (in milliseconds) for 10 key Gibbs sampler steps (corresponding to ψ, Υ, $\boldsymbol{\eta}_t$'s, $z_{jl}(\boldsymbol{s}_{i,o})$'s or $u_j(\boldsymbol{s}_{i,o})$'s, $\xi_j(\boldsymbol{s}_{i,o})$'s, θ_{jl_j}'s, $\delta_{1:k}$, ρ, κ, and $\alpha_{jl_j}(\boldsymbol{s}_{i,o})$'s) for our 4 models, i.e., `fullGPfixedL`, `NNGPblockFixedL`, `NNGPsequenFixedL`, and `NNGPsequenVaryLj`.

```r
setwd(wd)
load("GibbsStepTimeFixedLfullGP.RData")
load("GibbsStepTimeFixedLblock.RData")
load("GibbsStepTimeFixedLsequen.RData")
load("GibbsStepTimeVaryLjSequen.RData")
head(GibbsStepTimeFixedLfullGP, 50)
```

```
##         z  xi theta delta alpha kappa rho  eta upsilon psi
##  [1,] 125 131  2956    12    14    13  13 3080      21 503
##  [2,] 127 115  3018    12    14    16  13 3087      20 522
##  [3,] 128 123  2957    13    14    13  13 3028      21 511
##  [4,] 125 114  2922    13    14    13  13 3034      20 517
##  [5,] 132 111  2937    13    14    12  13 3185      22 591
##  [6,] 129 113  2919    13    14    13  13 3024      28 552
##  [7,] 125 120  2962    13    14    15  18 3023      31 545
##  [8,] 131 119  2984    12    14    15  18 3035      33 548
##  [9,] 133 118  3021    12    14    15  18 3054      32 556
## [10,] 126 111  2937    12    14    14  12 3079      21 533
## [11,] 126 135  2978    13    14    13  13 3153      20 545
## [12,] 131 163  3001    13    15    13  13 3036      21 554
## [13,] 130 112  2956    13    14    13  13 2999      20 514
## [14,] 127 110  2926    13    14    13  13 2976      21 516
## [15,] 125 111  2910    13    14    13  13 3024      21 520
## [16,] 129 110  2962    13    14    13  13 3020      21 528
## [17,] 126 111  2938    12    14    13  13 2998      20 509
## [18,] 125 112  2960    12    14    13  13 3053      22 521
## [19,] 132 112  3004    13    15    13  13 3043      21 519
## [20,] 128 117  3003    13    14    13  13 3087      21 517
```

[1] It is worth noting that we utilized a version of our package with the extra step-wise time recording code portions to fit the above models. This package version is significantly slower than our original package version presented in the GitHub repository.

```
## [21,] 130 115  2941    12    15    13  13 2985      21 511
## [22,] 131 127  2962    13    15    13  13 3051      20 518
## [23,] 127 111  2873    13    14    13  13 2956      21 499
## [24,] 127 151  2893    12    14    13  13 2977      20 542
## [25,] 124 138  2911    12    14    13  13 3006      20 543
## [26,] 133 114  2938    12    14    15  13 3023      20 538
## [27,] 126 113  2974    12    14    13  13 3056      21 513
## [28,] 125 115  2984    13    14    13  14 3067      20 511
## [29,] 122 118  2975    13    14    13  13 3077      20 507
## [30,] 120 110  3043    13    15    13  13 3080      20 507
## [31,] 120 115  2961    12    14    13  13 3133      20 514
## [32,] 118 114  2947    12    14    13  13 3013      20 508
## [33,] 120 117  2926    12    14    12  13 2984      20 533
## [34,] 126 120  2904    12    14    13  13 3156      22 553
## [35,] 121 110  2911    12    14    12  13 2981      21 511
## [36,] 129 117  2952    12    14    13  13 2969      20 511
## [37,] 125 110  2940    12    14    13  13 3017      21 516
## [38,] 126 112  2986    12    14    13  13 3058      21 513
## [39,] 125 115  2966    13    15    13  13 3155      21 565
## [40,] 125 110  2939    12    14    12  13 3002      21 513
## [41,] 125 110  2895    12    14    12  13 3015      20 515
## [42,] 123 110  2884    12    14    12  13 2993      20 507
## [43,] 127 113  2915    12    14    12  13 3026      20 504
## [44,] 126 112  2933    12    14    12  13 2982      20 499
## [45,] 126 111  2901    12    14    12  13 3006      20 516
## [46,] 124 112  2928    12    14    13  13 3010      24 547
## [47,] 129 128  2880    12    15    13  13 3009      20 543
## [48,] 125 110  2908    12    14    16  13 2986      21 537
## [49,] 122 144  2896    12    14    13  13 2994      20 545
## [50,] 128 145  2926    12    14    13  13 2996      20 542
```

```r
head(GibbsStepTimeFixedLblock, 50)
```

```
##         z  xi theta delta alpha kappa rho  eta upsilon psi
##  [1,] 130 117  2925    12    14    14  13 3059      20 529
##  [2,] 130 121  2991    12    14    14  13 3046      21 532
##  [3,] 136 123  2975    12    14    15  13 2998      21 540
##  [4,] 133 112  2919    12    14    13  13 3044      20 507
##  [5,] 131 111  2938    12    14    12  13 3017      20 509
##  [6,] 133 110  2925    12    14    12  13 3040      21 505
##  [7,] 130 110  2964    12    14    13  13 3031      20 501
##  [8,] 139 110  2958    12    14    13  13 3072      22 512
##  [9,] 138 113  3072    12    14    13  13 3066      21 508
## [10,] 132 111  2991    13    15    13  14 3103      21 535
## [11,] 133 113  2913    12    14    12  13 3028      23 535
## [12,] 130 111  2975    12    14    13  13 3022      21 506
## [13,] 142 111  2984    13    15    13  14 2988      21 510
## [14,] 132 112  2933    13    14    13  13 2986      21 513
## [15,] 129 112  2914    13    14    13  13 3030      21 520
## [16,] 130 115  2972    13    14    12  13 3041      21 516
## [17,] 130 114  2941    13    14    13  13 3012      20 526
## [18,] 132 117  2944    12    14    12  13 3055      21 505
## [19,] 134 125  2987    13    14    16  13 3030      22 527
## [20,] 134 142  2989    13    15    13  13 3115      22 563
## [21,] 144 118  2971    13    15    16  14 3020      27 559
```

```
## [22,] 123 140  2936    13    14    13  13 2981      21 556
## [23,] 125 148  2864    12    14    13  13 2930      21 548
## [24,] 122 138  2845    18    14    16  30 2947      34 571
## [25,] 124 120  2877    12    14    13  13 2977      22 532
## [26,] 127 112  2945    13    14    23  20 3054      33 566
## [27,] 123 122  2939    13    14    16  13 3045      34 568
## [28,] 134 124  2959    13    15    16  13 3020      21 531
## [29,] 131 119  2978    13    14    13  13 3060      20 514
## [30,] 131 113  3041    12    14    12  12 3067      21 525
## [31,] 130 119  2950    12    14    13  13 3113      22 532
## [32,] 135 124  2994    13    15    13  14 3177      22 556
## [33,] 131 131  2933    13    14    13  13 2979      21 540
## [34,] 130 150  2977    12    14    13  13 2960      20 549
## [35,] 130 146  2936    13    14    13  13 2993      21 548
## [36,] 127 148  2956    12    14    12  12 2956      21 557
## [37,] 141 148  2938    13    15    13  13 3030      21 543
## [38,] 133 159  2962    13    15    13  13 3009      20 539
## [39,] 135 161  2998    13    15    13  13 3065      21 568
## [40,] 135 150  2916    13    14    12  13 2973      20 541
## [41,] 129 117  2913    13    14    13  13 3025      21 518
## [42,] 131 118  2869    12    14    12  12 2977      20 507
## [43,] 129 110  2840    12    14    12  13 2980      20 506
## [44,] 132 111  2850    12    14    13  13 2933      20 498
## [45,] 131 109  2849    12    14    13  13 2981      20 507
## [46,] 130 114  2917    12    14    13  13 2987      20 500
## [47,] 136 116  2910    13    14    13  13 3015      20 515
## [48,] 130 109  2905    12    14    13  13 3022      20 513
## [49,] 133 111  2931    13    14    13  13 3030      21 523
## [50,] 131 110  2922    12    14    13  13 3000      21 517
```

```r
head(GibbsStepTimeFixedLsequen, 50)
```

```
##         z  xi theta delta alpha kappa rho  eta upsilon psi
##  [1,] 114 114  3002    13    21    15  13 3057      21 540
##  [2,] 112 112  2957    13    20    15  13 3073      21 553
##  [3,] 113 112  2946    12    20    15  13 3015      21 539
##  [4,] 119 110  2943    12    20    16  13 3027      20 533
##  [5,] 113 109  2921    12    20    16  13 2956      20 537
##  [6,] 118 110  2882    12    20    16  13 2955      20 538
##  [7,] 120 113  2907    12    20    15  13 2999      20 541
##  [8,] 121 112  2902    12    20    13  13 3010      21 508
##  [9,] 122 109  2946    12    20    12  13 3006      20 505
## [10,] 119 112  2948    12    20    13  13 2986      20 492
## [11,] 122 112  2964    12    20    13  13 3065      20 494
## [12,] 119 113  3020    13    21    13  13 3102      22 517
## [13,] 120 119  2999    13    21    13  13 3085      21 502
## [14,] 121 113  3008    13    20    13  13 3082      21 513
## [15,] 122 115  2959    13    20    13  13 3047      20 512
## [16,] 123 111  2920    13    21    13  13 3006      21 508
## [17,] 121 110  2934    12    21    13  13 3025      20 513
## [18,] 121 116  2935    12    21    13  13 3028      21 519
## [19,] 120 110  2957    12    20    13  13 3050      22 511
## [20,] 121 112  3003    12    20    13  13 3050      21 512
## [21,] 123 112  3012    13    21    13  13 3102      21 505
## [22,] 117 110  3056    13    20    13  13 3081      21 512
```

```
## [23,] 118 111  2923    12    20    13  13 3104      22 548
## [24,] 118 113  2959    13    20    13  13 3027      21 514
## [25,] 118 112  2934    12    20    13  13 3000      21 510
## [26,] 121 121  2962    12    20    13  13 2994      21 517
## [27,] 121 111  3037    13    20    13  13 2987      21 517
## [28,] 120 111  2949    12    20    12  13 2965      20 512
## [29,] 129 110  2956    13    20    13  14 3043      21 513
## [30,] 122 110  2965    13    21    14  14 3029      20 516
## [31,] 122 111  2988    13    21    13  14 3048      21 512
## [32,] 117 112  2932    13    21    13  13 3011      21 504
## [33,] 119 116  2942    13    20    13  13 3042      22 522
## [34,] 119 113  2866    13    20    13  13 2979      21 521
## [35,] 120 113  2822    12    20    12  13 2958      20 505
## [36,] 122 113  2852    13    21    13  13 2935      21 531
## [37,] 124 114  2856    13    21    13  13 2963      21 521
## [38,] 121 113  2865    12    20    13  13 3003      21 505
## [39,] 120 113  2899    13    21    13  13 3004      21 513
## [40,] 120 112  2882    13    20    13  13 3013      21 514
## [41,] 264 114  2984    13    21    13  13 3072      22 535
## [42,] 115 115  2939    13    21    13  13 3019      23 555
## [43,] 115 117  2960    13    20    13  13 3018      20 519
## [44,] 113 112  2953    13    20    13  13 3001      21 519
## [45,] 116 117  2978    13    21    13  13 3035      21 514
## [46,] 115 110  2913    12    20    13  13 3003      21 506
## [47,] 116 111  2907    13    20    13  13 2980      20 509
## [48,] 124 113  2926    12    20    13  13 3005      21 514
## [49,] 118 116  2877    13    20    13  13 3047      22 510
## [50,] 120 112  2869    13    20    13  14 3048      22 515
```

```r
head(GibbsStepTimeVaryLjSequen, 50)
```

```
##        u xi theta delta alpha kappa rho  eta upsilon psi
##  [1,] 12 37  2940    12    37    13  13 3038      21 533
##  [2,] 12 37  2902    12    40    12  12 3038      20 515
##  [3,] 12 37  2933    12    40    13  13 3053      21 527
##  [4,] 12 41  2931    12    39    12  12 3040      20 513
##  [5,] 12 37  2968    12    37    12  13 3058      21 499
##  [6,] 12 37  2967    12    41    12  13 3044      21 508
##  [7,] 12 36  2972    13    40    13  13 3089      21 509
##  [8,] 12 45  2990    13    39    13  13 2978      20 529
##  [9,] 12 38  2943    13    38    12  13 3066      21 522
## [10,] 12 37  2956    13    42    12  13 3023      22 530
## [11,] 13 37  2902    12    40    12  13 3018      21 515
## [12,] 13 39  2935    12    38    12  13 3015      21 514
## [13,] 13 39  2966    12    40    12  13 3016      21 520
## [14,] 12 39  2967    12    41    13  13 3051      20 510
## [15,] 12 37  3003    13    41    12  13 3036      20 490
## [16,] 12 37  3014    13    39    13  13 3024      22 502
## [17,] 12 37  2944    12    41    13  13 2943      21 511
## [18,] 12 37  2826    13    42    12  13 3055      21 520
## [19,] 13 40  3002    12    39    12  12 3060      21 533
## [20,] 12 37  2945    12    37    13  13 3061      21 517
## [21,] 13 46  2965    13    41    13  13 3082      20 504
## [22,] 12 42  3023    13    40    13  13 3115      21 506
## [23,] 13 38  2997    13    39    12  13 3091      21 498
```

```
## [24,] 13 38  2955    12    37    13  13 3051      21 514
## [25,] 12 39  2919    13    41    12  13 2924      21 531
## [26,] 12 39  3024    13    42    12  13 3060      21 530
## [27,] 12 40  3003    12    39    12  13 3076      21 521
## [28,] 12 40  2971    12    37    12  13 3019      22 521
## [29,] 12 44  2957    13    42    13  13 3059      20 519
## [30,] 13 43  3003    13    40    13  13 3086      21 515
## [31,] 12 38  2988    12    38    13  13 3056      20 498
## [32,] 12 38  2984    12    40    13  13 3051      20 490
## [33,] 13 38  2974    12    42    13  13 2997      20 503
## [34,] 13 38  2940    12    40    12  13 2985      20 518
## [35,] 13 43  2877    13    38    13  13 3038      22 565
## [36,] 12 39  2820    13    42    14  13 3038      21 532
## [37,] 12 37  3050    13    41    13  13 3021      21 530
## [38,] 12 37  2890    12    38    12  12 3050      20 520
## [39,] 12 39  2920    12    37    12  13 3040      20 487
## [40,] 12 37  2935    12    41    13  13 3027      21 502
## [41,] 12 46  2998    12    39    12  12 2983      21 512
## [42,] 12 37  2830    13    39    13  13 2939      20 518
## [43,] 12 38  3002    13    38    13  13 3077      21 514
## [44,] 12 37  3044    12    41    12  13 3018      21 523
## [45,] 12 36  2957    12    40    12  13 3045      21 528
## [46,] 12 46  2992    12    39    12  13 3089      20 530
## [47,] 13 41  2997    12    45    13  13 3089      21 507
## [48,] 12 37  3038    12    40    13  13 3078      21 501
## [49,] 12 38  3021    13    40    13  13 2979      20 503
## [50,] 13 38  2981    12    38    12  13 2959      22 507
```

We then present vital posterior sampling time summary statistics for the 10 key parameters.

```r
round(apply(GibbsStepTimeFixedLfullGP, 2, summary), 2)
```

```
##              z     xi   theta delta alpha kappa   rho     eta upsilon    psi
## Min.    114.00 106.00 2712.00 11.00 13.00 12.00 12.00 2774.00   19.00 475.00
## 1st Qu. 124.00 112.00 2891.00 12.00 14.00 12.00 13.00 2974.00   20.00 505.00
## Median  126.00 114.00 2926.50 12.00 14.00 13.00 13.00 3008.00   21.00 513.00
## Mean    128.04 115.96 2929.54 12.38 14.18 13.04 13.15 3011.56   21.06 517.59
## 3rd Qu. 129.00 117.00 2964.00 13.00 14.00 13.00 13.00 3044.00   21.00 528.00
## Max.    286.00 165.00 3165.00 28.00 31.00 34.00 45.00 3263.00   47.00 604.00
```

```r
round(apply(GibbsStepTimeFixedLblock, 2, summary), 2)
```

```
##              z     xi   theta delta alpha kappa   rho     eta upsilon    psi
## Min.    111.00 107.00 2741.00 11.00 13.00 12.00 12.00 2819.00   19.00 473.00
## 1st Qu. 124.00 112.00 2884.00 12.00 14.00 12.00 13.00 2972.00   20.00 507.00
## Median  128.00 114.00 2921.00 12.00 14.00 13.00 13.00 3006.00   21.00 518.00
## Mean    129.45 116.85 2923.71 12.61 14.35 13.65 13.33 3009.43   21.13 522.28
## 3rd Qu. 131.00 119.00 2960.00 13.00 14.00 14.00 13.00 3043.00   21.00 537.00
## Max.    280.00 161.00 3193.00 30.00 27.00 28.00 43.00 3279.00   48.00 614.00
```

```r
round(apply(GibbsStepTimeFixedLsequen, 2, summary), 2)
```

```
##              z     xi   theta delta alpha kappa   rho     eta upsilon    psi
## Min.    112.00 106.00 2704.00 12.00 19.00 12.00 12.00 2747.00   19.00 463.00
## 1st Qu. 123.00 111.00 2882.00 12.00 20.00 13.00 13.00 2972.75   20.00 508.00
## Median  126.00 114.00 2917.00 12.00 20.00 13.00 13.00 3006.00   21.00 523.00
## Mean    128.39 118.12 2920.04 13.77 21.58 15.05 15.06 3010.67   22.46 527.44
```

```
## 3rd Qu. 130.00 122.00 2955.00 13.00 21.00 16.00 13.00 3045.00   21.00 543.00
## Max.    294.00 166.00 3187.00 29.00 36.00 37.00 38.00 3258.00   43.00 618.00
```

```r
round(apply(GibbsStepTimeVaryLjSequen, 2, summary), 2)
```

```
##             u    xi   theta delta  alpha kappa   rho     eta upsilon    psi
## Min.    11.00 34.00 2695.00 11.00  35.00 11.00 11.00 2798.00   19.00 466.00
## 1st Qu. 12.00 37.00 2902.00 12.00  38.00 12.00 13.00 2986.00   20.00 498.00
## Median  12.00 38.00 2939.00 12.00  40.00 12.00 13.00 3024.00   21.00 508.00
## Mean    12.26 38.31 2937.98 12.34  40.08 12.49 12.81 3020.22   20.62 508.71
## 3rd Qu. 12.00 39.00 2976.00 13.00  41.00 13.00 13.00 3058.00   21.00 518.00
## Max.    16.00 57.00 3173.00 15.00 184.00 16.00 15.00 3271.00   25.00 585.00
```

```r
round(apply(GibbsStepTimeFixedLfullGP, 2, sd), 3)
```

```
##       z      xi   theta   delta   alpha   kappa     rho     eta upsilon     psi
##  16.505   7.541  55.614   0.771   0.927   1.491   1.601  53.926   2.521  17.848
```

```r
round(apply(GibbsStepTimeFixedLblock, 2, sd), 3)
```

```
##       z      xi   theta   delta   alpha   kappa     rho     eta upsilon     psi
##  15.826   8.240  56.833   1.708   1.608   2.235   2.521  53.998   2.587  19.541
```

```r
round(apply(GibbsStepTimeFixedLsequen, 2, sd), 3)
```

```
##       z      xi   theta   delta   alpha   kappa     rho     eta upsilon     psi
##  16.288   9.934  55.152   3.833   3.867   4.334   5.573  55.426   4.809  23.018
```

```r
round(apply(GibbsStepTimeVaryLjSequen, 2, sd), 3)
```

```
##       u      xi   theta   delta   alpha   kappa     rho     eta upsilon     psi
##   0.478   2.608  58.438   0.479   8.906   0.528   0.421  58.706   0.627  15.162
```